

Sinchana K

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Professional Summary

Final-year Artificial Intelligence and Machine Learning undergraduate with strong foundations in Python, SQL, Data Structures and Algorithms, Object-Oriented Programming, and Machine Learning. Hands-on experience building AI-powered applications using RAG, LLMs, LangGraph, and Flask. Passionate about developing scalable software solutions, solving real-world problems, and continuously learning modern AI technologies.

Technical Skills

Languages: Python, SQL

Core CS: Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks

AI/ML & GenAI: Machine Learning, NLP, RAG, LLMs, Embeddings

Frameworks/Libraries: Pandas, NumPy, Streamlit, LangChain, LangGraph, Flask

Databases and Vector Stores: MySQL, PostgreSQL, FAISS

Tools/Platforms: Git, GitHub, Docker, VS Code, Ollama, Groq API

Projects

AI Research Agent | GitHub

Python, LangChain, LangGraph, Streamlit, Ollama, Phi-3, arXiv API, PyPDF2

- Designed and developed an AI-powered research agent using RAG and LangGraph for autonomous research analysis across multiple academic papers.
- Integrated arXiv API, LangChain, and Phi-3 via Ollama to build an end-to-end pipeline for contextual academic content generation.
- Engineered a Python-based multi-document PDF processing system using PyPDF2 to extract, aggregate, and analyze 5,000+ words of academic text from research papers.
- Built a Streamlit-based interactive interface that displays the top 3 relevant research paper links along with AI-generated summaries and research reports.

Hybrid Adaptive RAG Query System | GitHub

Python, FAISS, HuggingFace, Llama 3, Groq API

- Developed a hybrid system that dynamically switches between Retrieval-Augmented Generation (RAG) and LLM responses for improved contextual accuracy.
- Implemented semantic search using FAISS and HuggingFace embeddings to retrieve relevant document context for precise answer generation.
- Integrated Llama 3 via Groq API, achieving low-latency responses suitable for real-time applications.

Boston House Price Prediction System | GitHub

Python, Pandas, NumPy, Scikit-Learn, Flask, Docker

- Built an end-to-end machine learning application to predict housing prices using 506 housing records and 13 property-related features.
- Performed exploratory data analysis, correlation analysis, train-test splitting, feature standardization, and model training using Pandas, NumPy, and Scikit-Learn.
- Deployed the trained Linear Regression model through a Flask web application and containerized the solution using Docker for portable execution across environments.

Education

Bachelor of Engineering (B.E) in Artificial Intelligence & Machine Learning

2023 – 2027

BMS College of Engineering, Bengaluru

CGPA: 8.31 / 10.0

Certifications

- Accenture Software Engineering Job Simulation – Forage
- GenAI for Professionals: 10X Your Productivity – Udemy